

**Highlight**

Database research in crystal engineering

Ashwini Nangia*

*School of Chemistry, University of Hyderabad, Hyderabad 500 046, India.
E-mail: ansc@uohyd.ernet.in*

*Received 4th February 2002, Accepted 22nd February 2002
Published on the Web 28th February 2002*

The Cambridge Structural Database is a storehouse of over 250 000 small-molecule organic and organometallic crystal structures. Statistical analysis of structural data provides a better understanding of the nature of hydrogen bonds and intermolecular interactions and facilitates in the identification of frequently occurring interaction patterns and supramolecular synthons. With an enhanced knowledge of hydrogen bond synthons in crystals, novel strategies may be designed for the self-assembly of supramolecular architectures from multi-functional molecules. The twin phenomena of polymorphism and pseudopolymorphism have been studied by extracting information on molecular conformations and solvent inclusion from the database. The approach geometry of hydrogen bond donors to various functional groups derived from small-molecule crystal structures is used as a model in structure-based drug design and pharmacophore mapping. Thus, the implications of crystal engineering today go far beyond organic and metal–organic crystal design into supramolecular materials, nanotechnology, ligand–protein binding and crystal structure prediction. This Highlight surveys the importance of database research in some of these areas.

Introduction

The term ‘crystal engineering’ was first suggested by Schmidt¹ in the 1970s in connection with his pioneering work on the topochemical reactions of crystalline cinnamic acids. The meaning of the term was widened in the 1990s by Desiraju:² “Crystal engineering is defined as the understanding of intermolecular interactions in the context of crystal packing and in the utilisation of such understanding in the design of new solids with desired physical and chemical properties”. The subject started with the study of organic solids. Today crystal engineering deals with the self-assembly of both organic and inorganic systems: molecular crystals,³ nanostructures,⁴ metal–organic frameworks,⁵ and coordination polymers.⁶ The nature of intermolecular interactions that mediate molecular recognition in these two types of systems is the same for both: strong O–H...O, N–H...O, O–H...N hydrogen bonds;⁷ weak C–H...O, C–H...N hydrogen bonds;⁸ heteroatom interactions X...X,⁹ X...O¹⁰ (X = Br, I); π – π interactions; and van der Waals forces. In coordination polymers, strong metal–ligand bonding is used in conjunction with hydrogen bonding to build crystalline architectures. Steiner has discussed the nature and properties of hydrogen bonds in the solid state in a recent review.¹¹ A proper understanding of the strength and

directional preferences of hydrogen bonds and intermolecular interactions lies at the heart of controlling the target crystal structure. Since the energy scale of heteroatom interactions and hydrogen bonds (0.5–15 kcal mol^{−1}) is roughly an order of magnitude weaker than covalent bonds, the former can be easily deformed by packing forces and hence exhibit variable geometry in crystals. There are three techniques to study hydrogen bonding: X-ray crystallography, spectroscopy and computation. This article deals with the study of intermolecular interactions through statistical analysis of geometrical information obtained from a large number of crystal structures. An advantage with the database approach is that any distortion in individual interaction geometry is averaged out in the diverse sample of X-ray crystal structures determined for a variety of reasons. The reader may refer to other reviews on this topic.¹²

The Cambridge Structural Database¹³ (CSD, www.ccdc.cam.ac.uk) is a storehouse of X-ray crystallographic data on organic and organometallic molecules. Starting from less than 2000 entries in 1965, the CSD today contains accurate X-ray crystal structures of over a quarter of a million compounds. The CSD is growing at the rate of about 10% every year, is updated biannually, and operates with Unix and Windows (ConQuest) interface. The CSD has built-in, user-friendly data retrieval (QUEST3D), statistical analysis (VISTA), and crystal structure display and visualisation (RPluto, Mercury) packages. The output from a particular query may be compared with the knowledge base of intermolecular interactions (IsoStar). In conjunction with add-on programs like GOLD (protein–ligand docking), Relibase+ (searching for protein–ligand complexes), DASH (structure solution from powder diffraction data) and DBUse (database of publications that use the CSD), chemists and crystallographers have convenient access to a large amount of crystallographic data and structural information.

In this review, I will highlight some important advances published in the period 1999–2001 on the use of the Cambridge database: (1) to understand the nature of intermolecular interactions; (2) to study the conformation and arrangement of molecules in the crystal; (3) to extract repetitive structural patterns in crystals; (4) to develop new crystal engineering strategies; and (5) to understand polymorphism and pseudo-polymorphism. Hydrogen bond motifs extracted from the CSD

Ashwini Nangia received his M.Sc. degree at the Indian Institute of Technology, Kanpur, in 1983. He then worked with Frederick Ziegler on the total synthesis of natural products and obtained his Ph.D. from Yale University in 1988. He joined the University of Hyderabad in 1989 and is now Reader in the School of Chemistry. His current research interests are in the area of crystal engineering and supramolecular chemistry. He has published over 80 papers on non-covalent synthesis, host–guest chemistry, hydrogen bonding and database research. For more information, see <http://www.uohyd.ernet.in>.



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provide useful inputs in structure-based drug design and molecular modelling. Programs such as GOLD and Relibase+ establish a link between small-molecule and macromolecular crystallography (PDB, Protein Data Bank). I will touch upon this interface to show how molecular conformations and interaction geometries in the CSD are useful for ligand modelling and drug–receptor docking simulations. It is not possible to cover the numerous papers published on database research in this short Highlight. Interested readers may download more citations through the DBUse facility of the CCDC.

At the outset I recapitulate some standard practices in database work. Searches must be carried out on accurate crystal structures (R factor <0.1), with atom coordinates present, a chemical/crystallographic connectivity match, being error-free, and with no disorder. While defining the geometry of any hydrogen bond, $D-H\cdots A$, the $D-H$ distance must be normalised to the neutron diffraction value (HNORM). This is because the H-atom position is systematically underestimated by X-ray diffraction. The $H\cdots A$ distance, d , should be in the range 1.5–2.2 Å for strong $O-H\cdots O/N$ hydrogen bonds and 2.0–3.0 Å for weak $C-H\cdots O/N$ contacts. The $D-H\cdots A$ angle, θ , should be defined in the range 140–180° for strong hydrogen bonds and 120–180° for weak interactions. The distance–angle and bit screen criteria may be altered depending on the specific application and/or the number of hits retrieved. Angular distributions must be corrected for the difference in solid angle sweep of the donor hydrogen as a function of θ with the weighting factor $1/\sin \theta$, the so-called cone correction (Fig. 1). The reader may refer to the excellent monograph by Desiraju and Steiner⁸ for a detailed discussion of these points.

Historical

The first paper on the statistical analysis of weak intermolecular interactions, $C-H\cdots O$, $C-H\cdots N$ and $C-H\cdots Cl$, appeared in 1982. Taylor and Kennard¹⁴ provided crystallographic evidence for the existence of these weak interactions from a survey of 113 neutron diffraction crystal structures retrieved from the CSD. A few years later, it was shown that halogen $\cdots$ halogen contacts are of two types, type I (triclinic) and type II (monoclinic).¹⁵ Significantly, only type II halogen $\cdots$ halogen contacts are polarisation-induced, namely with electrophile–nucleophile pairing ($C-X^{\delta+}\cdots X^{\delta-}$). Type I contacts are space filling. The significance of donor acidity in $C-H\cdots O$ hydrogen bonds was examined by comparison of the $C\cdots O$ distance histogram for Cl_3CH , Cl_2CCH and ClC_2CH donors.¹⁶ A pK_a scale of carbon acidity was developed by plotting the mean $H\cdots O$ distance in the range 3.0–4.0 Å for donors with $C-H$ acidity in the pK_a range 10–50.¹⁷ $C-H$ donors with low pK_a form short hydrogen bonds whereas less acidic donors have longer contacts. All this naturally led to studies on the basicity of the acceptor group. Mean $C-H\cdots O$ bond distances increase in the order $P=O < C=O < C-O-C < M-CO$ when the donor acidity is held constant, reflecting the change in acceptor strength.¹⁸ In organometallic structures ($M-CO$ acceptor), the mean $C\cdots O$ distance is longer for terminal CO than for bridged CO.¹⁹ The chemical reason for this difference is that bridged CO is more basic than terminal CO because of efficient π back-donation from metal to ligand. These crystallographic studies together with spectroscopic measurements and computational data provided insight into the properties of

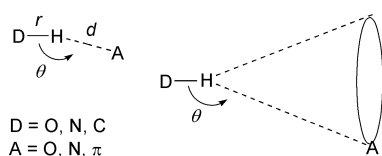


Fig. 1 Typical hydrogen bond, $D-H\cdots A$, and the cone correction, $1/\sin \theta$.

weak $C-H\cdots O$ hydrogen bonds in crystals: (1) the $H\cdots O$ distance and $C-H\cdots O$ angle range from 2.2–2.8 Å and 120–180°, with the best interactions being short *and* linear; (2) these interactions are electrostatic in nature, being stabilised by $C^{\delta-}-H^{\delta+}\cdots O^{\delta-}$ charges; (3) their energy is in the range -0.5 to -4.0 kcal mol⁻¹; and (4) the strength of a $C-H\cdots O$ hydrogen bond increases with donor acidity and acceptor basicity, with the best combination being acidic donor *and* basic acceptor. In effect, $C-H\cdots O$ hydrogen bonds exhibit the characteristics of strong $O-H\cdots O$ hydrogen bonds (5–15 kcal mol⁻¹), except that they are weak.

The nature of the $C-H\cdots N$ interaction was called into question in a recent paper dealing with the crystal structure of bis(2,2'-pyridylamido)cobalt(II), $Co(dpa)_2$.²⁰ The authors postulated that the $C-H\cdots N$ interaction (d 2.44 Å, θ 177°) in their crystal structure is a 'real' hydrogen bond whereas numerous other examples are mere van der Waals contacts. This claim was immediately rebutted by two papers through statistical analysis of $C-H\cdots N$ and $C-H\cdots O$ interactions in the Cambridge database. That the $C-H\cdots N$ geometry in $Co(dpa)_2$ is not unique was easily proved by the retrieval of intermolecular $C-H\cdots N$ contacts in the range 2.05–2.45 Å, which showed that there are *ca.* 1000 observations with $d < 2.45$ Å and *ca.* 100 entries with θ in the range 170–180°.²¹ In another paper that immediately followed, it was shown that the directionality of $C-H\cdots O$ interactions for different types of $C-H$ donors decreases with decreasing polarisation of the donor atom.²² The angularity of $C-H\cdots O$ contacts decreases in the order $(sp)C-H (\approx O-H) > (sp^2)C-H > (sp^3)C-H$ because the ability of $C-H$ donors to form hydrogen bonds is related to their pK_a ($C\equiv C-H$ 25, $C=C-H$ 43). Since these angular distributions were reviewed recently,¹¹ they are not displayed here.

Intermolecular interactions

The nature of the $O-H\cdots O$ hydrogen bond changes with the $H\cdots O$ distance, d . The interaction is mostly electrostatic when d is >1.6 Å with very little charge transfer component. As d is reduced (<1.5 Å), the interaction becomes quasi-covalent and is like a half-covalent bond when the proton is centred between the O-atoms ($r \approx d$). Furthermore, as the $H\cdots O$ distance d becomes shorter, the $O-H$ bond r lengthens in classical hydrogen bonds. Inter-anion $O-H\cdots O$ hydrogen bonds are frequently observed in crystal structures that contain HSO_4^- , $H_2PO_4^-$, HPO_4^{2-} (and other inorganic anions) and oxalate, fumarate, and squarate (in general carboxylate) anions. Based on 125 hydrogen bond geometries extracted from 66 ordered and error-free neutron crystal structures, it has been shown that the covalent $O-H$ bond lengthens as the $H\cdots O$ hydrogen bond becomes shorter for inter-anion cases like $O^+-H\cdots O^-$, $O^+-H\cdots O^-$, $O-H\cdots O^-$ and even $O^--H\cdots O^-$ (Fig. 2).²³ When accurate hydrogen atom position is very important, as in this case, only neutron diffraction data can be trusted. While these situations have varying degrees of electrostatic and charge transfer components, they are all in the regime of the strong hydrogen bond model explained for the neutral $O-H\cdots O$ interaction.⁷ Inter-anionic $O-H\cdots O$ interactions are being examined by many research groups using X-ray diffraction, spectroscopy and computational methods.²⁴

The occurrence of ring and chain type motifs through $O-H\cdots O^-$ and $O^--H\cdots O^-$ interactions (Fig. 3) in crystal structures containing three common anionic groups was analysed: hydrogen carbonate (HCO_3^-), hydrogen oxalate ($HC_2O_4^-$) and hydrogen squarate ($HC_4O_4^-$).²⁵ Hydrogen carbonates prefer to form rings whereas hydrogen oxalates form chains only. Hydrogen squarates form both ring and chain motifs. Mean $O\cdots O$ distances in the 70 or so structures examined are in a narrow range of 2.50–2.60 Å.

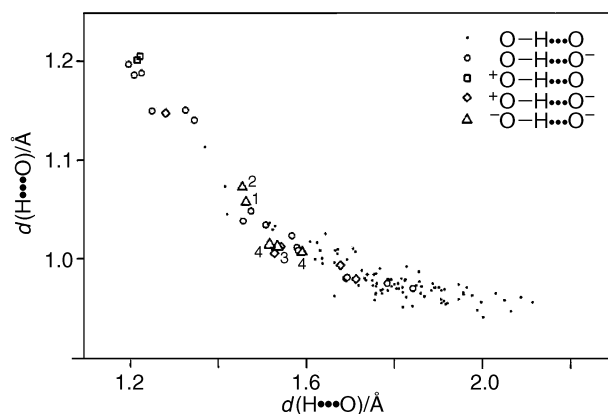


Fig. 2 Lengthening of the covalent O–H bond in O–H...O hydrogen bonds. Note that the points for the O[−]–H...O[−] interaction (Δ) follow the same trend as normal O–H...O bonds. Symbols 1, 2, 3, 4 correspond to compounds discussed in ref. 23. Reproduced from *Chem. Commun.*, 1999, 2299.²³

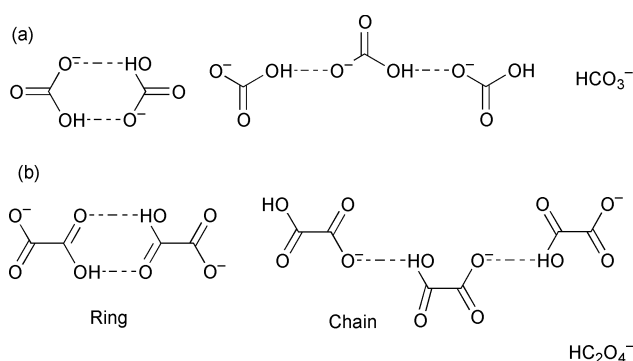


Fig. 3 Ring and chain motifs in hydrogen carbonate (a) and hydrogen oxalate (b) anions.

The rapid appearance of a number of papers highlighting the structural role of the C–H...Cl interaction encouraged us,²⁶ as well as others,²⁷ to examine the nature of this interaction in crystal structures archived in the database. The order of acceptor strength with O–H, N–H and C–H donors is Cl[−] (chloride ion) > Cl–M (metal-bound chlorine; M = transition metal) > Cl–C (organic chlorine). When the donor is strong (OH and NH groups), the H...Cl interaction exhibits clear characteristics of hydrogen bonds with basic acceptors (e.g. Cl[−] and Cl–M) and the behaviour persists even for neutral organic chlorine (Cl–C). However, in the case of weakly polarised C–H donors, the interaction exhibits weak hydrogen bond type behaviour for Cl[−] and Cl–M acceptors but C–H...Cl–C behaves more like a van der Waals contact. Thus, the transition from C–H...Cl–M to C–H...Cl–C represents the borderline between a weak hydrogen bond and a van der Waals contact. In addition to the histogram and scatter plot features discussed in the original paper,²⁶ the data in Table 1 support our conclusion. The mean C–H...Cl distance increases as the acceptor becomes less basic, from Cl[−] to Cl–M to Cl–C. This is

Table 1 Mean *d* and *θ* values for the C–H...Cl interaction, with (e.s.d) and [no. of observations] as indicated in parentheses

	C–H...Cl [−]	C–H...Cl–M	C–H...Cl–C
(all)C–H			
Mean <i>d</i> /Å	2.941(7) [1059]	2.974(5) [1424]	3.050(4) [1395]
Mean <i>θ</i> /°	136.14(59)	135.41(51)	133.93(51)
(sp ²)C–H			
Mean <i>d</i> /Å	2.876(9) [617]	2.941(8) [517]	3.051(4) [1469]
Mean <i>θ</i> /°	137.44(75)	135.13(84)	133.47(49)

as might be expected. Secondly, the mean distance is shorter in the subset of acidic (sp²)C–H donors compared with the (all) C–H category for Cl[−] and Cl–M acceptors but not for Cl–C. This suggests that there is minimal electrostatic contribution in the C–H...Cl–C interaction and hence it is termed as a van der Waals contact. It may be noted that analysis of geometrical data is difficult in situations such as this where the so-called ‘signal’ and the background ‘noise’ are of comparable magnitude. In another paper that appeared about the same time, Brammer *et al.*²⁸ concluded that the C–H...Cl–C interaction is a *very* weak hydrogen bond by plotting normalised distance and spatial distributions. Thus, the transition from very strong hydrogen bonds (15–40 kcal mol^{−1}) to strong hydrogen bonds (4–15 kcal mol^{−1}), and from strong hydrogen bonds to weak hydrogen bonds (0.5–4 kcal mol^{−1}) and van der Waals contacts is a smooth continuum. These results suggest that the C–H...Cl–C interaction is a borderline case at the bottom of the energy scale (0.5–2 kcal mol^{−1}).

So far I have discussed interactions in which the acceptor is a single atom. An acceptor can also be a multi-atom group, say the phenyl ring π-cloud. A natural question with a phenyl ring as the hydrogen bond acceptor is: does the donor H-atom point toward the ring centroid, or to a C=C bond midpoint, or to a particular C-atom? This issue has been recently addressed using the database approach.²⁹ Realising that a multi-atom acceptor group has a finite area, it was reasoned that hydrogen bond trajectory frequencies must be corrected by a geometrical factor before they become chemically meaningful. If the offset distance from the phenyl ring centroid to the projection of the donor H-atom on the phenyl ring plane is *R*, then an area correction 1/*R* must be applied to frequency distributions because the number of points is proportional to 2π*R*. When such an area correction is applied to O–H...π, N–H...π and C–H...π frequency distributions, then approach of the donor is clearly to the ring centroid and not to the bond midpoint (Fig. 4). The offset corrected plots (d), (e) and (f) clearly show that chemical recognition is preferred at the phenyl ring centroid and not toward the periphery of the ring as in histograms (a), (b) and (c). Approach of the donor hydrogen to the phenyl ring centroid is energetically favoured because in this way the electropositive hydrogen can simultaneously interact with six acceptor atoms. The *N/R* area correction for a multi-atom acceptor group complements the *N/sin θ* angular correction for single-atom acceptors.

Detailed methods for exploring C–H...π contacts in the database may be found in a study that establishes the role of this weak and soft interaction in controlling the conformation of peptides.³⁰ On the other hand, weak C–H...S interactions do not display structure-directing properties in crystals³¹ as shown by a CSD survey.

Interaction patterns

Individual hydrogen bonds and intermolecular interactions are by themselves weak. For this reason, the identification of robust multi-point recognition motifs or interaction patterns is often more profitable in crystal design. Hydrogen bond and intermolecular interaction motifs that occur repeatedly in a crystal structure are termed supramolecular synthons,³² and accordingly, crystal engineering may be viewed as a kind of supramolecular synthesis in the solid state.³³ The goal in crystal engineering therefore is to dissect the target supramolecular network in terms of the constituent molecules and the recurring (robust) supramolecular synthons. Thus, the chemist must plan recognition between molecular functional groups *via* robust hydrogen bond synthons for the design of predictable crystalline architectures. The available libraries of starting materials and reaction protocols afford convenient access to a large chemical diversity through molecular synthesis. The imagination and

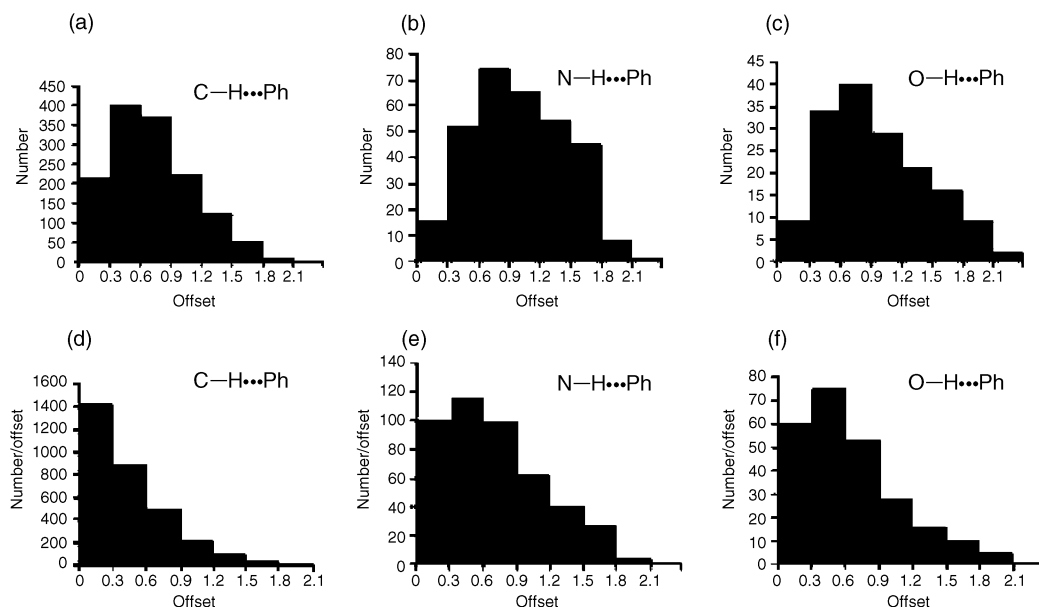


Fig. 4 Uncorrected (N vs. R) and corrected (N/R vs. R) histograms for $X-H\cdots\pi$ hydrogen bonds: (a), (b) and (c) are uncorrected plots, and (d), (e) and (f) are area corrected distributions. Note that approach to the ring centroid is highlighted in the corrected plots. See text for the definition of R . Reproduced from *Chem. Commun.*, 2001, 703.²⁹

ingenuity of the chemist is the only limiting factor in the design of novel supramolecular structures (nanostructures) given the tools of chemical synthesis and the rules of close packing and hydrogen bonding.

A hierarchy of supramolecular synthons was established by calculating the probability of occurrence of 75 common non-covalent ring motifs constructed from $O-H\cdots O$, $O-H\cdots N$, $N-H\cdots O$ and $N-H\cdots N$ hydrogen bonds.³⁴ It may be noted that recognition between unlike functional groups (*e.g.* melamine–barbital **I**, acid–aminopyridine **III**) has far greater probability of formation than synthons between identical functional groups (*e.g.* acid dimer **IV**, amide dimer **VI**) (Fig. 5). Database analysis shows that generally robust synthons with a high occurrence probability result from uncommon functional groups whereas synthons with lower probability arise from functional groups that are common in organic molecules. Can one find a recurring supramolecular synthon for a ubiquitous functional group? In this direction, structural data on recognition of carboxyl donor with complementary acceptor groups was analysed.³⁵ The carboxylic acid–pyridine synthon **VIII** through $O-H\cdots N$ and $C-H\cdots O$ hydrogen bonds is a recognition motif between unlike functional groups that has a success rate of over 90% in crystals. These studies provide a strong argument in favour of synthesising tailored molecules with functional groups grafted in proper positions on the molecular scaffold for systematic crystal design. This is illustrated through pyrazinecarboxylic acids that crystallise *via* the recurring acid–pyridine synthon **VIII**.³⁶

The occurrence of ring motifs for weak $C-H\cdots O$ hydrogen bonded synthons assembled through two- and three-point recognition was examined.³⁷ An advantage in practising supramolecular synthesis with weak interactions is that the available library of functional groups expands many-fold (*e.g.* NO_2 , CN , $C\equiv CH$, halogen, phenyl) compared with strong hydrogen bonds only (OH , NH_2 , CO_2H , $CONH_2$). In this respect, 1,3,5-trinitrobenzene has emerged as a popular choice in the crystal engineering of molecular complexes and layered materials.³⁷ Halogen bonding is a new paradigm that complements hydrogen bonding for self-assembly.³⁸ The halogen trimer synthon was identified as a recurring pattern in crystals and exploited in the construction of hexagonal microporous solids that include guest molecules in the voids.³⁹ To summarise, supramolecular synthesis may be implemented on a wide

palette of molecular diversity by using a judicious combination of strong and weak hydrogen bonding functional groups.

Recognition between phenyl groups is of three main types: π – π stacking in the face-to-face and offset-slipped orientations, and the edge-to-face herringbone T-motif.⁴⁰ While the perfectly aligned π -stacked motif is mostly stabilised by quadrupole–quadrupole interactions between σ - and π -frameworks, the parallel displaced and herringbone T-geometry are stabilised by electrostatic $C-H^{\delta+}\cdots\pi^{\delta-}$ forces. A number of intermediate

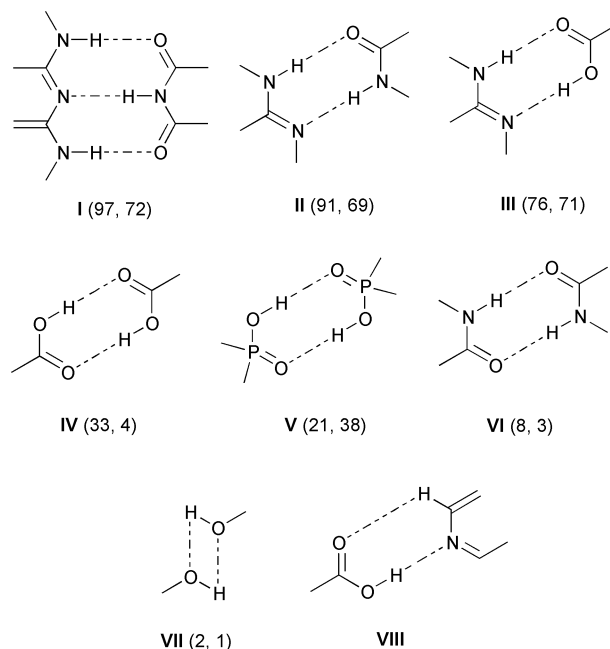


Fig. 5 Some common cyclic hydrogen bond synthons **I–VII** taken from ref. 34. The probability of formation, P_m (%), and the number of structures that contain the component functional groups, S_{poss} (rank), is given in parentheses. P_m is in the range 0.4–97% (rare to frequently occurring synthons) and the S_{poss} rank is in the range 1–75 (frequent to rare functional groups). Note that when P_m is high (robust synthon), the S_{poss} rank is low (uncommon functional group) and *vice versa*. The acid–pyridine synthon **VIII** combines the dual features of a common functional group with a robust supramolecular synthon for crystal engineering.

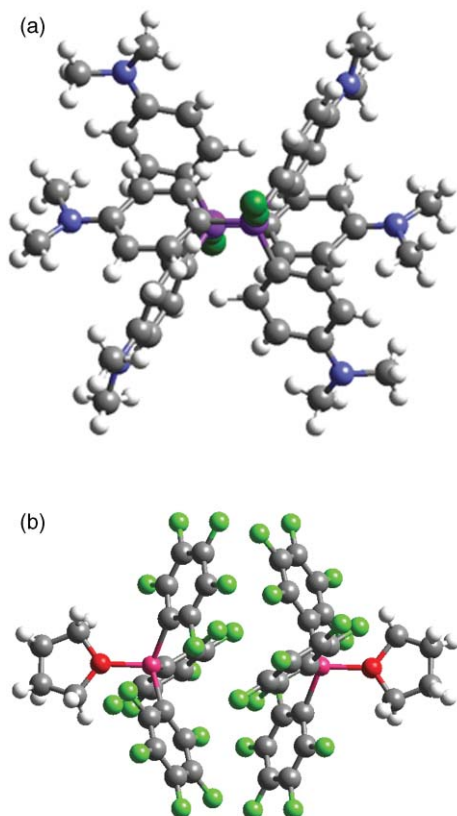


Fig. 6 (a) Six-fold phenyl embrace in $[\{4-(\text{CH}_3)_2\text{NC}_6\text{H}_4\}_3\text{P}]\text{AuCl}$ ($\text{P}\cdots\text{P} = 6.78 \text{ \AA}$) viewed along the pseudo three-fold axis of the 6PE. Note that the large 4-NMe₂ substituents overhang the periphery of the embrace and do not interfere with its formation. (b) Centrosymmetric perfluorophenyl embrace 6P^FE in $(\text{thf})\text{Al}(\text{C}_6\text{F}_5)_3$ ($\text{Al}\cdots\text{Al} = 6.17 \text{ \AA}$). Note the topological similarity between the supramolecular motifs of perfluorinated and hydroaromatic molecules. Reproduced from *J. Chem. Soc., Dalton Trans.*, 2000, 2909⁴¹ and *New J. Chem.*, 2000, 24, 295.⁴³

situations are also possible that optimise electrostatic stabilisation between the positively charged hydrogen and the negatively charged π -cloud. Given that these numerous recognition motifs are weak ($0.5\text{--}2.0 \text{ kcal mol}^{-1}$), can one use phenyl groups to engineer molecular crystals? In the context of organometallic crystals it has been shown that structures with Ph_4P^+ cations and PPh_3 ligands associate through non-bonded phenyl–phenyl interactions that are attractive, concerted and widespread.⁴¹ The strongest and most prevalent interaction in the database is the sextuple phenyl embrace (6PE, energy $10\text{--}13 \text{ kcal mol}^{-1}$) in which there are six concerted edge-to-face (eff) interactions [Fig. 6(a)]. The 6PE is a robust motif, being present in *ca.* 30% of crystal structures with Ph_4P^+ cations. Substitution of small groups, like methyl, does not perturb the embrace motif; *ortho*- and *meta*-positions are more sensitive than *para*-substitution. Moreover, the multiple phenyl embrace motifs persist in XPh_3 -containing crystal structures even in high symmetry cubic and rhombohedral lattices.⁴² The occurrence of similar aggregation patterns in fluorinated aromatic molecules is also demonstrated.⁴³ There are examples of the six-fold perfluorophenyl embrace (6P^FE) in the database [Fig. 6(b)] and the energy of this motif is estimated to be $5\text{--}12 \text{ kcal mol}^{-1}$. Four-fold phenyl embrace supramolecular motifs, 4PE and 4P^FE, have also been noted in crystal structures.

Database studies have been used to trace similarities between the directionality and interaction patterns of $\text{O}\cdots\text{H}\cdots\text{O}$ hydrogen bonds and ‘secondary bonds’, namely those involving heavy p-block elements such as Bi(III) and chloride ions. The 3c-4e σ -bonded interaction $\text{Cl}\cdots\text{Bi}\cdots\text{Cl}$ in $[\text{BiCl}_2\text{Ph}_2]^-$, formed by the addition of Cl^- to BiPh_2Cl , has a strong preference for the linear approach akin to $\text{O}\cdots\text{H}\cdots\text{O}$ hydrogen bonds [Fig. 7(a) and (b)].⁴⁴ The Lewis base Cl^- donates an electron pair to the $\text{Bi}\cdots\text{Cl}$ σ^* orbital in a near-linear approach such that the $\text{Cl}\cdots\text{Bi}\cdots\text{Cl}$ moiety occupies the diaxial position in the trigonal bipyramidal structure. The ring and chain networks of $\text{Cl}\cdots\text{Bi}\cdots\text{Cl}$ bonds [Fig. 7(c)] in ER_2X type crystal structures [$\text{E} = \text{Lewis acid: Bi(III), Sb(III), I(III), Te(IV)}$; $\text{X} = \text{Lewis base: halide, alkoxide}$] are topologically identical to $\text{O}\cdots\text{H}\cdots\text{O}$ hydrogen bond networks in crystalline monoalcohols.

Hydrogen bond and intermolecular interaction patterns

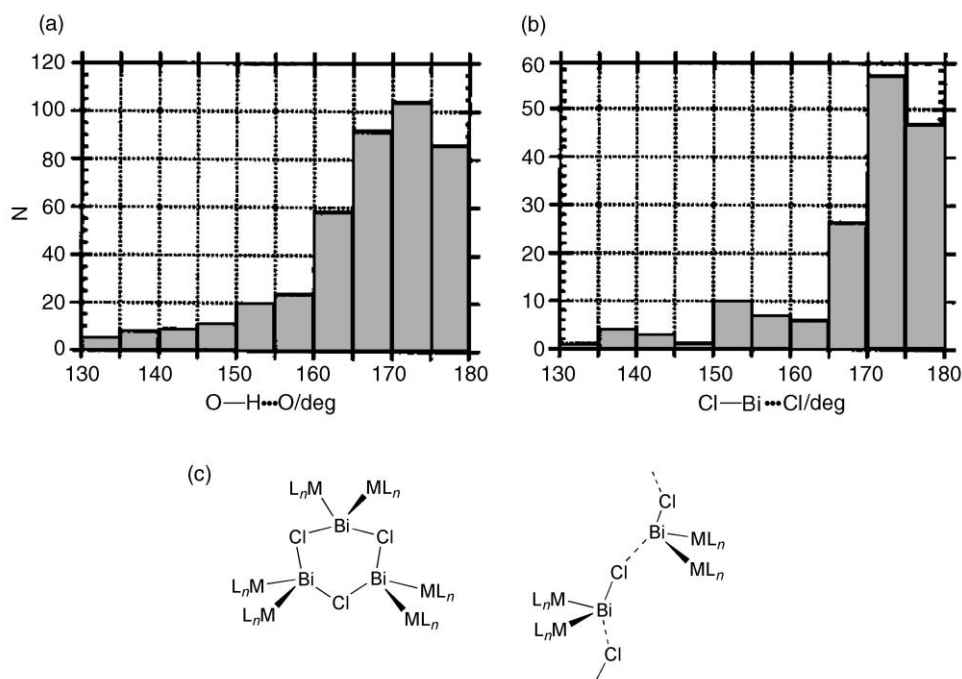


Fig. 7 Histograms of angles (°) in (a) $\text{O}\cdots\text{H}\cdots\text{O}$ and (b) $\text{Cl}\cdots\text{Bi}\cdots\text{Cl}$ systems. (c) Typical ring and chain motifs in ER_2X species. Reproduced from *New J. Chem.*, 1999, 23, 969.⁴⁴

may be represented mathematically using the graph set notation.⁴⁵ In this notation, a crystal structure is encoded as a G-array and represented as a constructor graph with the molecules represented as vertices and the directed hydrogen bonds, $H\cdots A$, as $>H$ -edges. For example, hydrogen bond patterns in the polymorphs of iminodiacetic acid and the method to write the corresponding constructor graphs are described.⁴⁶ Algorithms have been developed for the automatic assignment of graph set descriptors in PLUTO to facilitate visualisation of structures and retrieval of data.⁴⁷

Crystal packing

The frequency of space groups and number of molecules in the asymmetric crystal unit, Z' , have interested crystallographers, especially when the space group is rare or the Z' value is high or low. Data were retrieved on 51 different values of Z' covering the range from 1/96 to 32.⁴⁸ Many of these values occurred only rarely: 29 were flagged less than ten times, 19 only once or twice. The trends are better revealed in the top ten Z' values summarised in Table 2 which account for over 99% of the database. There are strong substance related dependencies in the occurrence of $Z' > 1$. The average frequency of $Z' > 1$ in organic compounds is 10.8%, in steroids 18.8%, and for nucleosides/nucleotides 20.8%. In general, high Z' values (e.g. 12 and 16 in hydrated trimesic acid and cholesterol, respectively) could result from one or more of these factors: a space-filling problem of awkwardly shaped molecules in simple close packing modes; the presence of multiple low energy conformations; or because strong hydrogen bonded aggregates crystallise from solution without reorganisation into a symmetrical array in the solid state. On the other hand, crystals with very low Z' values (e.g. 1/48 and 1/96 in the plastic phase of adamantane and C_{60} at room temperature, respectively) are examples of disordered structures. An attempt was made in this paper to trace the onset of the higher frequency of crystal structures with very high Z' values with the recent change-over world wide to CCD detectors for X-ray data collection. There is no apparent increase, but this could be either because the technical specifications for published crystal structures have improved in terms of low R factor or because it is too early to see discernible shifts.

In a database study that closely followed the above one, conformational similarity between crystallographically independent enantiomers in 114 non-centrosymmetric racemates was quantified using the r.m.s. deviation in molecular superposition.⁴⁹ Unfortunately, there is no automated method to search the CSD for racemic compounds. The authors therefore searched a small but randomly selected part of the database manually (5560 against 50291 entries) and concluded that racemates constitute 23% of the centrosymmetric organic structures. Among the 27695 non-centrosymmetric organic

Table 2 The ten most frequent Z' values in organic and organometallic crystal structures

Rank	Organic		Organometallic	
	Z'	Relative frequency (%)	Z'	Relative frequency (%)
1	1	73.3(2)	1	62.4(2)
2	1/2	13.7(1)	1/2	26.2(1)
3	2	9.6(1)	2	5.67(7)
4	3/4	1.00(3)	1/4	2.34(5)
5	3	0.49(2)	1/3	0.92(3)
6	4	0.47(2)	1/6	0.89(3)
7	1/6	0.42(2)	1/8	0.45(2)
8	1/3	0.34(2)	1 1/12	0.26(2)
9	1 1/12	0.22(2)	3	0.25(2)
10	1/8	0.18(2)	4	0.17(1)
Sum of top 10		99.57(2)		99.53(2)

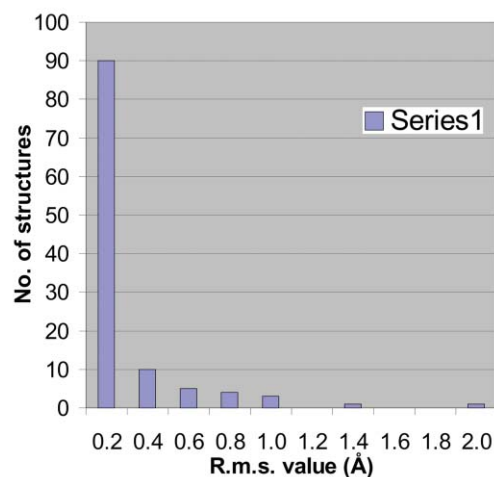


Fig. 8 R.m.s. deviation in atomic coordinates of heavy atoms between pairs of crystallographic independent molecules in 114 non-centrosymmetric organic racemates.

crystals, 3819 structures have glide symmetry. In these 3819 structures, 632 racemates were identified and a majority (93.3%) belong to one of the five space groups, Pc (6.2%), Cc (21.5%), $Pca2_1$ (23.3%), $Pna2_1$ (36.9%) and $Fdd2$ (5.5%). The r.m.s. deviation histogram from the overlay of atomic coordinates in 114 non-centrosymmetric racemates is displayed in Fig. 8. The average r.m.s. deviation (non-H atoms only) for the data in Fig. 8 is 0.19 Å, with almost 80% of the structures having r.m.s. deviation < 0.2 Å. The corresponding numbers for crystallographically independent molecules in chiral space groups is 0.32 Å and 55%. Thus, crystallographically independent molecules of opposite chirality are conformationally more similar compared with molecules with the same chirality.

Crystal density and packing variations were compared among crystalline isomers.⁵⁰ The spread of density is smaller for polymorphs than for isomers with different constitution and molecular shape. Polycyclic cage molecules have a higher density than condensed aromatics, which in turn are denser than flexible molecules. The maximum density difference is 20%, with about 60% of cases having a density difference of $< 3\%$. A possible relationship between hydrogen bonding and crystal density, namely that strong intermolecular interactions should lead to a compact crystal, was analysed through spot checks but was not found to be operative. A reason for the lack of any correlation could be that the strongly directional property of hydrogen bonding can also produce open frameworks (e.g. ice) and the two factors presumably cancel each other. High crystal density does not correlate with higher lattice energy, even though intuitively one may have expected such a trend. In the subset of 164 hydrocarbon structures, the mean packing fraction is 0.721(22), in effect confirming that efficient crystal packing follows the cubic closest packing of spheres model.

Polymorphism and pseudopolymorphism

Polymorphism may be defined as the existence of the same chemical substance in more than one crystalline modification.⁵¹ Pseudopolymorphism is the phenomenon of solvent inclusion in crystals. A systematic understanding of crystal packing and crystal lattice energies is possible through polymorphs because these crystal structures represent systems with a minimum number of variables. Getting the correct polymorph is of the utmost importance in drugs, pharmaceuticals, dyes, pigments, confectionery products and supramolecular materials. Pseudopolymorphism is the phenomenon wherein a compound is obtained in crystalline forms that differ in the nature or

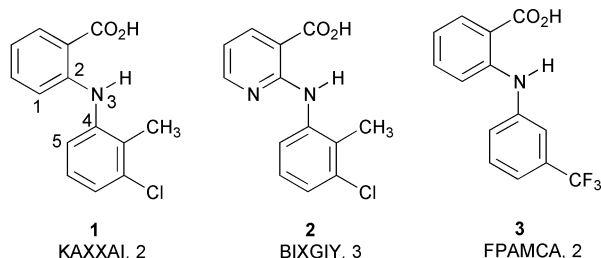


Fig. 9 Polymorphism in diarylamines 1–3. The CSD refcodes and number of forms are mentioned. Torsion angle τ_1 = atoms 2–3–4–5, and τ_2 = atoms 1–2–3–4.

stoichiometry of included solvent. Pseudopolymorphs are distinct chemical and structural entities. This fact may be exploited in patenting and litigation of active drug forms.

Polymorphism is known in diarylamine type drug molecules, tolfenamic acid **1**, the nicotinic acid derivative **2**, and flufenamic acid **3** (Fig. 9).⁵² These molecules are similar in structure but have different numbers of polymorphs. The polymorphs differ in torsion angles, τ , about the two C–N bonds, and hence they are classified as conformational polymorphs. In order to compare the conformation of aryl rings in these polymorphs, data on torsion angles in related molecules were plotted. Most of the compounds have τ_2 in the narrow range of -10 to 20° . τ_1 values occur in a wider range of 0 – 120° . Based on this analysis, some new polymorphs are predicted to be more stable than those archived in the CSD.

Polymorphism is closely related to ‘crystal structure prediction’ since whenever polymorphs are found, the natural question is: are there other forms yet to be discovered? *Ab initio* crystal structure prediction⁵³ is the major challenge in crystal engineering and the Cambridge database provides crucial structural inputs essential for solving this problem. Programs like the Polymorph Predictor (*Cerius*)⁵⁴ rank thousands of putative structures of a given molecule in a narrow energy range of *ca.* 1 kcal mol^{-1} . The question as to which of these is the correct structure cannot be decided based on enthalpy considerations alone because of the role of entropy during crystallisation. The kinetic information on crystallisation is encapsulated in the quarter of a million or so crystal structures in the CSD, a number that is expected to double by the end of this decade. In a very recent paper, Sarma and Desiraju⁵⁵ have used chemical arguments in conjunction with the molecular diversity that is available in the CSD to eliminate unrealistic structures from the output of *ca.* 1000 low energy frames calculated by the Polymorph Predictor. They have correctly predicted that the crystal structure of imide **4** contains the catemer synthon **X** and not the dimer synthon **IX** through a computational-cum-database method (Fig. 10).

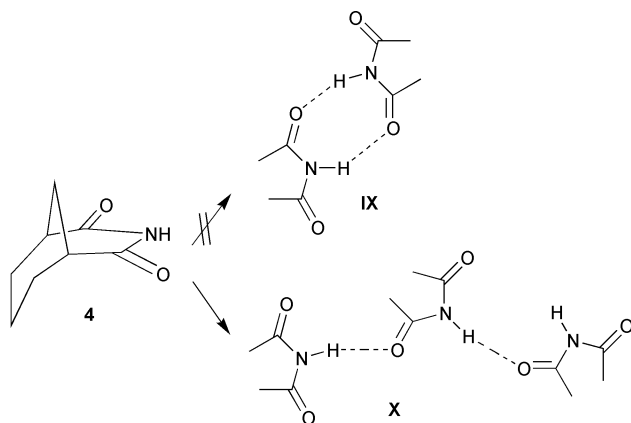


Fig. 10 Crystal structure prediction using the molecule $\rightarrow$ synthon $\rightarrow$ structure approach.

How far such an approach like this will be valid in a general sense still remains to be seen but the addition of chemical intuition into what has hitherto been a purely computational exercise is surely expected to yield dividends.

In an attempt to better understand pseudopolymorphism, the occurrence of 20 common solvents that appear at least 50 times in organic crystal structures was retrieved from the CSD.⁵⁶ These solvents were ranked in the order of their occurrence, O_{corr} . The fact that different solvents are used to different extents by chemists for crystallisation was normalised by correcting for solvent usage through a manual inspection of *Acta Crystallographica, Section C* volumes of 1986 and 1996. Thus, a value of 1.00 for O_{corr} indicates that the solvent is included in organic crystals commensurate with its usage. Values much greater and much less than unity mean that these solvents have greatly enhanced and greatly reduced tendency to form solvates, respectively. Dimethylformamide (DMF), dimethyl sulfoxide (DMSO) and dioxane are the top solvents (O_{corr} 5.69, 4.73 and 4.70, respectively), and EtOH, Et₂O and *n*-hexane are least likely to be included in crystals (O_{corr} 0.46, 0.35, 0.11, respectively). A model based on multi-point hydrogen bonding between solute and solvent molecules is proposed to explain the formation of pseudopolymorphs, which account for 15% of organic crystals in the CSD. A more exhaustive study with 50 solvents appeared subsequently.⁵⁷

Structure-based drug design

It was pointed out almost two decades ago⁵⁸ that the geometrical data on intermolecular contacts in small-molecule crystal structures could be used to construct ligand–receptor recognition models. An early program, HSITE,⁵⁹ uses hydrogen bond frequencies in the CSD to predict hydrogen bonding sites in proteins. An advantage in using empirical propensities of recognition motifs compared to their interaction energies is that competition between different functional groups, as may be present in actual systems, is accounted for in the former method. There are other advantages with structural data derived from the CSD even though small-molecule crystal structures may not be as relevant to *in vivo* systems as the PDB macromolecular structures. (1) The CSD data are of much higher accuracy and the diversity of functional groups is larger; (2) CSD scatter plots are based on larger numbers of observations and so the trends and outliers are clear; (3) H-atom positions are known for a majority of the CSD structures and so contouring on the hydrogen atom in the probe nucleus is possible. A library of interaction sites in proteins, SuperStar, has been developed from non-bonded contacts in the CSD (IsoStar) and validated with 122 X-ray structures of protein–ligand complexes.⁶⁰ Propensity maps are generated for different probes: NH_3^+ nitrogen, carbonyl oxygen, hydroxyl oxygen, and methyl carbon atoms. Peaks in the SuperStar map of different probes may be used as an interaction blueprint of the binding site. These propensity blueprint maps serve as a guide in the manual docking of ligand into the enzyme active site for structure-based drug design and 3D pharmacophore development.

A two-parameter structural model for biologically active β -lactam antibiotics was proposed based on the geometry of β -lactam molecules in the Cambridge database.⁶¹ In the 30 penicillins, which constitute a majority of active β -lactams, the ‘closed’ (axial CO_2H) conformation is stabilised by a combination of intramolecular $\text{N–H}\cdots\text{S}$ and $\text{C–H}\cdots\text{O}$ hydrogen bonds, estimated to be *ca.* 3 kcal mol^{-1} .⁶² On the other hand, the less active or inactive penicillin sulfoxides and sulfones (19 entries) adopt the ‘open’ (equatorial CO_2H) conformation. Thus, a connection is established between the axial- CO_2H conformation, stabilisation through hydrogen bonding, and biological activity in penicillin antibiotics.

The known inhibitors of the *Leishmania* DHFR (dihydrofolate reductase) enzyme contain the 2,4-diaminopyrimidine fragment. A new structural motif might improve the selectivity and pharmacokinetic properties. Novel inhibitors of *Leishmania* DHFR were screened that have a completely different basic chemical structure.⁶³ Using the active site of the enzyme as input, the CSD was searched with the DOCK3.5 program and the compounds scored with the built-in force field. From 86 compounds, five molecules with different skeletons and low scores were selected for biological assays (the lower the score, the better the predicted ligand–enzyme interaction energy). These ligands inhibit the enzyme in μM concentration.

Conformation of the drug molecule plays an important role in binding to the receptor protein. However, the preferred conformation of candidate molecules may not be known in a docking simulation. For example, binding of the sulfonamide inhibitor **5** is *ca.* 10-fold better than the carboxamide inhibitor **6**. In order to find out the most likely conformation of **5**, distribution of C–N–S–C torsion angles in the C–NH–S(O₂)–C substructure was plotted for both small-molecule (CSD) and ligand–protein complex (RELIBase) crystal structures.⁶⁴ The mean value of 184 occurrences in the CSD is $68.8 \pm 11.5^\circ$ and 15 ligands in the PDB is $71.1 \pm 11.6^\circ$. Based on the bent geometry about the sulfonamide group, compared to the *trans* conformation of the carboxamide group, docking of these inhibitors into the Zn–metalloproteinase active site showed that interface between the biphenyl substituent and the protein is much larger for sulfonamide **5** compared with carboxamide **6** (399 ± 13 vs. $204 \pm 70 \text{ \AA}^2$) (Fig. 11). Thus, the superior binding of inhibitor **5** is explained based on hydrogen bonding and hydrophobic interactions, with a crucial experimental input for the molecular dynamics simulation being derived from the CSD.

The exchange of H with an F-atom is an attractive approach for the design of fluorinated drug molecules and enzyme inhibitors. In this context, 50 pairs of organic CH₃/CF₃ crystal structures were manually retrieved from the CSD and their packing characteristics compared.⁶⁵ Exchange of CH₃ with CF₃ does not result in major structural differences when the molecule is large/rigid (>15 carbon atoms, tricyclic) and/or when the structure is stabilised by robust supramolecular synthons (*e.g.* synthons **I** and **IV** in Fig. 5). Steroids combine both these structural features and hence are excellent systems to

test CH₃/CF₃ exchange mimicry ideas. The dominant isostructural category has 24 pairs, about half the structures analysed. This predictive model for H/F exchange has applications in crystal engineering, drug design and materials science.

Conclusions

The Cambridge Structural Database is an indispensable research tool in crystal engineering, supramolecular chemistry and structure-based drug design. It is an accurate storehouse of structural information on chemical bonding, molecular conformations, hydrogen bonding and intermolecular interactions. Database studies have advanced in the past two decades from the study of hydrogen bonding to the identification of recurring hydrogen bond patterns and molecular conformation of active drug forms. Database research is intimately linked to crystallographic experiments and *vice versa*. For example, occurrence of the weak C–H $\cdots$ Cl interaction in some organometallic crystals led to detailed statistical analysis on the nature of this interaction. In the retrosynthesis sense, the robust acid–pyridine supramolecular synthon extracted from the database was exploited to construct predictable supramolecular assemblies. In effect, each layer of results set the stage for future ideas in the pyramid of scientific research. The subject of crystal engineering started with solid-state topochemical reactions over two decades ago and continues to elicit interest from supramolecular chemists.⁶⁶ Today, research areas under the wide umbrella of crystal engineering include: nanotechnology; supramolecular synthesis; separation science and catalysis; supramolecular materials and devices; polymorphism and crystal structure prediction; molecular modelling; and drug design and ligand–protein binding.⁶⁷

Certain questions that are somewhat difficult to address at the present time will be tackled with greater confidence as the database grows and the methods to mine the information become more refined. For example, to understand the border between a very weak hydrogen bond and a van der Waals contact, to correlate crystal structure with function and property (*e.g.* frequency doubling, melting point), and to select the correct structure from the thousands of putative frames generated by crystal structure prediction programs. All this means exciting challenges not only for chemical research but also for software development.

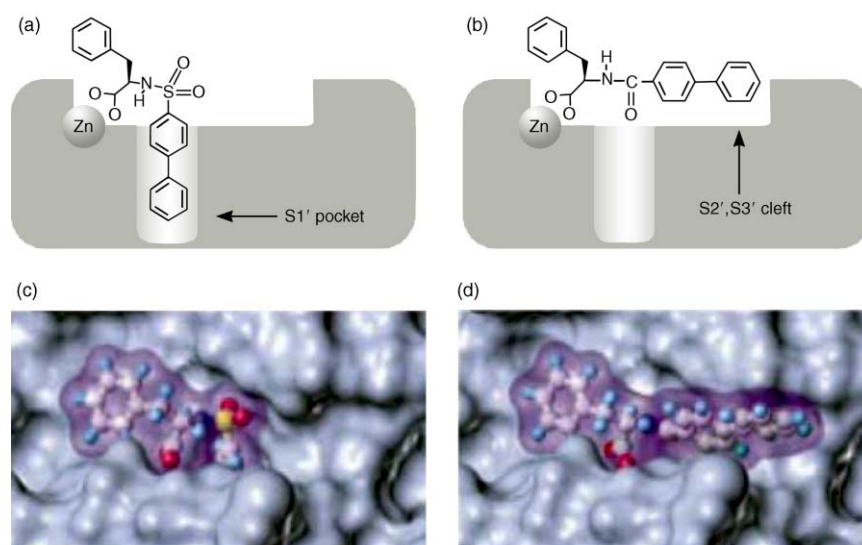


Fig. 11 Schematic model for the docking of inhibitors **5** and **6** into the Zn–metalloproteinase enzyme (a, b). Note that in sulfonamide **5** the C–N–S(O₂)–C torsion adopts a *gauche* conformation and the biphenyl group plunges into the deep S1' pocket (c). In carboxamide **6**, the C–N–C(O)–C torsion is in a *trans* conformation and the biphenyl substituent is directed to the shallow S2' and S3' cleft (d). Reprinted with permission from *J. Med. Chem.*, 1999, **42**, 1723.⁶⁴ © 1999 American Chemical Society.

Acknowledgements

The author thanks the Department of Science and Technology, Government of India, for research funding (Project No. SP/S1/G-29/98).

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